Leo Chao

leo.yt.chao@gmail.com | linkedin.com/in/leoytchao | twleochao.github.io

Education

University of California, San Diego | La Jolla, CA *Sep 2026 - Present*
Master of Science, Computer Science and Engineering

University of California, Irvine | Irvine, CA *Sep 2023 - Jun 2026*
Bachelor of Science, Computer Science - Intelligent Systems (GPA: 3.96/4.00)

Honors: Outstanding Contribution to Research (Faculty Nominated), Dean's Honors List, Accelerated 3-Year Degree

Skills

Programming Languages: Python, C++

LLM & Agents: Google ADK, Vertex AI, PyTorch, HuggingFace Transformers, LangGraph, LLM-as-a-Judge pipelines

Interp & Eval: TransformerLens, SAELens, Activation Patching, Attribution Methods, Sparse Autoencoders (SAE)

Tools: GCP, Vertex AI, Docker, Git, Linux

Research Experience

Julian McAuley Lab - UC San Diego | La Jolla, CA, *Research Collaborator* *May 2026 - Present*
- Collaborating with Xin Xu on mechanistic interpretability of music LLMs using sparse autoencoders and TransformerLens

ARCALA Research Lab - UC Irvine | Irvine, CA, *Research Assistant* *Jan 2025 - Jun 2026*
- Co-developed Chat-Bloom, an LLM evaluation taxonomy derived from Bloom's taxonomy, to classify and measure behavioral shifts in human-LLM interaction logs; organized and analyzed a dataset of 8,076 prompt-response pairs
- Trained BERT/DistilBERT ensemble classifiers for automated interaction taxonomy labeling, achieving ∼90% accuracy validated against human-coder baseline (75.9% inter-rater reliability)
- **Chao, L.**, et al. "Chat-Bloom: A Taxonomy for Classifying Cognitive Offloading in Student-AI Interactions in CS Education." Accepted, SIGCSE Virtual 2026

Professional Experience

LSEV USA Corp | Irvine, CA, *AI Intern* *Jun - Aug 2026*
- Designed a quality filter that decides which LLM-extracted product claims stay published, using cross-family judging and source re-grounding, reaching 96% agreement with human labels across a 300-product set labeled by two annotators
- Reduced published claim errors by 65% across 2,444 product pages by splitting validation into separate evidence-existence and sentiment-correctness checks
- Built a 3-layer query classifier resolving 80% of product searches deterministically under 40ms at zero API cost, cutting LLM spend by 82% versus judging every query

Acer | Taipei, Taiwan, *AI Systems Intern* *Jun - Aug 2025*
- Architected a multi-agent SDLC automation system using Google ADK and Vertex AI, reducing manual iteration time by 35%; deployed to GCP with a production path into internal Google Agentspace
- Refactored LangGraph prototype into production-grade ADK workflow with persistent memory management, integrating 6 enterprise GCP services including Cloud Storage and Firestore
- Designed state management architecture for long-horizon agent tasks, ensuring context preservation across 10+ dynamic subtask agents executing code generation, API stub creation, and test drafting

Projects

Reward Model Interpretability | github.com/twleochao/rm-attribution-patching *Jul - Sep 2026*
- Reproduced a published negative result on 8B reward-model interpretability at 100x the original scale, measuring attribution/patching rank correlation at -0.276 $[-0.316, -0.236]$
- Built two causal controls showing early-layer patching measures reward fragility and not information content: unrelated donors disturbed reward more than rejected answers in 84/100 pairs, with no advantage at matched disturbance
- Found and corrected a completeness bug in the authors' attribution library that omitted 47-58% of the model's reward.